SHAP Explainability Framework

Predictive Model and Explainability Analysis

Model Performance

An XGBoost classification model was developed to predict employee attrition using the left variable as the target. The model demonstrates exceptionally strong predictive performance, with an overall accuracy of 97.77% and an ROC AUC of 0.9927, indicating a very high ability to distinguish between employees who leave and those who remain. Performance metrics for the attrition class further support the model's reliability, with precision of 96.95% and recall of 93.56%, resulting in an F1 score of 0.9522. The confusion matrix indicates a low number of misclassifications relative to the total sample size, confirming that the model is highly effective at identifying employees at risk of leaving. Overall, these results demonstrate that the model provides a strong and reliable foundation for analyzing the key drivers of employee turnover.

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Accuracy: 0.9777
Precision: 0.9695
Recall: 0.9356
F1 Score: 0.9522
ROC AUC: 0.9927

Classification Report:

              precision    recall  f1-score   support

           0       0.98      0.99      0.99      2286
           1       0.97      0.94      0.95       714

    accuracy                           0.98      3000
   macro avg       0.97      0.96      0.97      3000
weighted avg       0.98      0.98      0.98      3000

Confusion Matrix:
[[2265   21]
 [  46  668]]
```

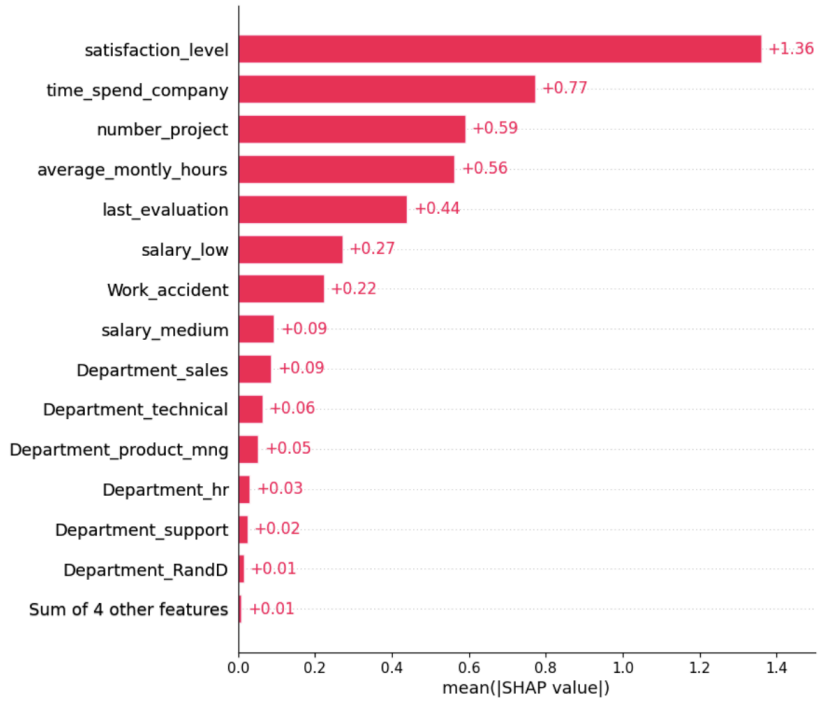
Explainability Framework (SHAP Analysis)

To interpret the model's predictions, SHAP (SHapley Additive exPlanations) was applied as an explainability framework. SHAP values quantify the contribution of each feature to both overall model behavior and individual predictions, allowing for a transparent understanding of the factors influencing employee attrition.

The SHAP analysis provides both:

- Global explanations, identifying the most important drivers of turnover across the dataset
- Local explanations, illustrating how specific employee characteristics contribute to individual attrition risk

Key Drivers of Employee Turnover



	feature	importance
0	satisfaction_level	0.250259
4	time_spend_company	0.232068
2	number_project	0.145059
1	last_evaluation	0.095107
3	average_montly_hours	0.076466
5	Work_accident	0.042862
16	salary_low	0.026756
17	salary_medium	0.019389
10	Department_management	0.016983
15	Department_technical	0.015125
9	Department_hr	0.014260
14	Department_support	0.013738
12	Department_product_mng	0.012850
13	Department_sales	0.012401
6	promotion_last_5years	0.009098

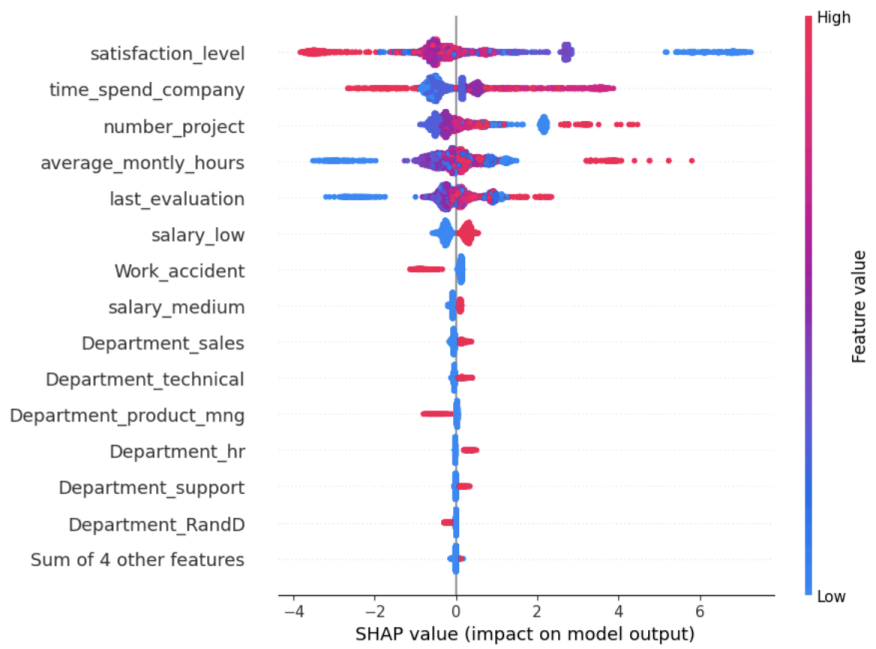
The SHAP summary and feature importance results indicate that employee attrition is primarily driven by a small number of key variables:

- Satisfaction Level is the most significant predictor of turnover. Lower satisfaction levels strongly increase the likelihood of employee departure, making it the dominant driver across the model.
- Time Spent at Company plays a major role, indicating that attrition risk varies across tenure levels. This aligns with earlier findings showing elevated turnover among mid-tenure employees.
- Number of Projects is a critical factor, with extreme project assignments—particularly very high workloads—significantly increasing attrition risk.

- Average Monthly Hours reinforces the importance of workload, as employees working excessive hours exhibit a higher probability of leaving.
- Last Evaluation Score also contributes meaningfully to turnover predictions, suggesting that high-performing employees may be at risk due to increased expectations or pressure.

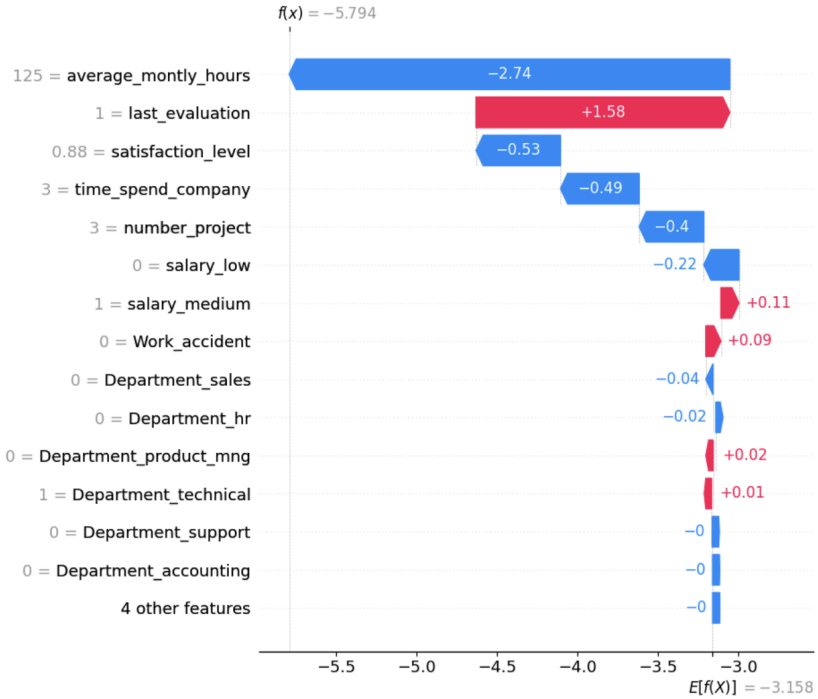
Secondary variables such as salary level and work accidents have smaller but measurable effects, while department-level factors show minimal impact, indicating that attrition is more strongly driven by individual-level characteristics than organizational structure.

SHAP Summary Interpretation



The SHAP beeswarm plot highlights both the magnitude and direction of each feature's impact on turnover. Lower satisfaction levels consistently push predictions toward employee departure, while higher project counts and longer working hours also increase attrition risk. Importantly, the model captures non-linear relationships, particularly in workload-related variables. Both extremely high and, in some cases, very low workload levels contribute to higher turnover risk, suggesting that imbalance in work allocation is a key driver of employee exits.

High-Risk Employee Profile



To further understand attrition patterns, the model was used to identify employees with the highest predicted probability of leaving. Analysis of these high-risk individuals reveals a highly consistent and actionable profile.

The employees most at risk of attrition exhibit:

- Very low satisfaction levels (0.09–0.11)
- High performance evaluations (0.80–0.90)
- Maximum project load (7 projects)
- Extremely high working hours (296–310 hours per month)
- Mid-level tenure (approximately 4–5 years)
- No recent promotions (promotion_last_5years = 0)

Additionally, these employees span multiple departments, indicating that this pattern is not isolated to a specific organizational unit but is instead a broader structural issue. The model assigns these employees predicted attrition probabilities above 99.9%, and all individuals in this group have already left the organization, confirming the model's accuracy and reinforcing the validity of the identified pattern.

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       satisfaction_level  number_project  average_montly_hours  \
340                  0.09             7.0                 308.0
12331                0.11             7.0                 302.0
14542                0.11             7.0                 302.0
12517                0.09             7.0                 309.0
1153                 0.10             7.0                 300.0
1228                 0.10             7.0                 310.0
559                  0.11             7.0                 296.0
1963                 0.09             7.0                 310.0
654                  0.10             7.0                 308.0
614                  0.09             7.0                 296.0

       time_spend_company  promotion_last_5years  predicted_probability_left
340                   5.0                    0.0                    0.999929
12331                 4.0                    0.0                    0.999922
14542                 4.0                    0.0                    0.999922
12517                 4.0                    0.0                    0.999915
1153                  4.0                    0.0                    0.999912
1228                  5.0                    0.0                    0.999908
559                   4.0                    0.0                    0.999908
1963                  4.0                    0.0                    0.999905
654                   4.0                    0.0                    0.999901
614                   4.0                    0.0                    0.999901
```

Key Insight

A critical insight from this analysis is that:

The employees most at risk of leaving are not low performers, but rather high-performing, overworked, and dissatisfied employees.

This indicates that attrition is driven less by poor performance and more by workload imbalance, burnout, and lack of engagement among valuable employees.

Alignment with Prior Analysis

The findings from the SHAP explainability framework strongly reinforce the earlier descriptive analysis:

- High turnover among specific project levels is confirmed by the importance of number_project
- The role of workload imbalance is supported by both average_monthly_hours and project-related findings
- Elevated turnover among mid-tenure employees aligns with the significance of time_spend_company
- The relationship between salary and retention is reflected, though it is less influential than workload and satisfaction

This consistency across analytical methods strengthens the overall reliability of the conclusions.

Business Implications

The analysis highlights a clear and actionable retention challenge. While overall turnover may appear manageable, attrition is concentrated among a critical group of employees who are both highly productive and heavily utilized.

This suggests that the organization may be inadvertently losing valuable talent due to:

- Excessive workload and project assignment
- Insufficient attention to employee satisfaction
- Limited career progression opportunities

Addressing these factors presents a significant opportunity to improve retention outcomes and reduce the loss of high-performing employees.